Thomas Y.L. Lin

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EDUCATION

University of Washington — Seattle, WA, USA
B.S. in Physics, Minor in Applied Math; GPA: 3.95 — 2025 – 2027 (expected)

National Taiwan University — Taipei, Taiwan
B.S. in Physics, B.A. in Economics — 2023 – 2025 (transferred)
Honors: Dean's List Award (Physics Dept., 1st Rank, Fall 23 & Spring 25); BaBar-SURF Award (one of 8 awardees)

EXPERIENCE

Swallowtail Labs — San Francisco, CA, USA
Co-Founder & CEO — May 2026 – Present
- Building a multi-physics (electrical-thermal-optical) surrogate model on chips for planning their own thermodynamics.
- Partnered with MediaTek after a 90-min pitch to the Chairman, securing angel funding on the research prototype.

California Institute of Technology — Pasadena, CA, USA
Summer Undergraduate Research Fellowships (SURF) — March 2025 – January 2026
- Developed Decoupled Diffusion Inverse Solver (DDIS), a physics-aware, data-efficient generative framework for PDE inverse problems, decoupling coefficient prior learning from physics enforcement.
- Proved a guidance-attenuation failure mode in prior joint-embedding models under limited data, justifying our design.
- Achieved SOTA with 1% paired data, improving reconstruction error by 11% (L2) and 54% (spectral) over prior best solvers on inverse Helmholtz, Poisson, and Navier–Stokes problems.
- Advised by Prof. Anima Anandkumar and PhD student Jiachen Yao.

Northwestern University — Remote
Machine Learning Intern — April 2023 – April 2025
- Analyzed the fine-grained complexity of modern Hopfield models; designed approximation algorithms reducing time complexity from subquadratic to near-linear (ICML 2024).
- Built a latent Bayesian network for portfolio optimization — a data-driven Black-Litterman model learning investor views as latent variables (ICML 2025).
- Advised by Prof. Han Liu and PhD candidate Jerry Yao-Chieh Hu.

Gamma Paradigm Capital — Taipei, Taiwan
Quantitative Analyst Intern — June 2023 – July 2024
- Implemented Distributionally Robust Optimization portfolio strategies (Blanchet, Chen, and Zhou, 2022) across 11 sectors (2006–2023): Sharpe [0.68, 0.78], MDD [-18.4%, -27.1%] vs. SPY (0.55, -55.2%) and EQW (0.54, -37.2%).
- Built a data-driven Black-Litterman model: +49.8% Sharpe over Markowitz (0.66–0.87 vs. 0.35–0.62) and S&P 500 / DJIA on 20- and 30-year data; 55.1% lower turnover, robust to hyperparams. Spun out into research with Northwestern.

PUBLICATIONS (**Equal Contribution*)

On JEPA Isotropy — **NeurIPS 2026**
Thomas Y.L. Lin, Han Liu, Jerry Yao-Chieh Hu. — Manuscript Link

Decoupled Diffusion Sampling for Inverse Problems on Function Spaces — **AI&PDE @ ICLR 2026 (Oral)**
Thomas Y.L. Lin*, Jiachen Yao*, Lufang Chiang, Julius Berner, Anima Anandkumar. — ArXiv Link

Latent Variable Estimation in Bayesian Black-Litterman Models — **ICML 2025**
Thomas Y.L. Lin, Jerry Yao-Chieh Hu, Paul W. Chiou, Peter Lin. — ArXiv Link

On Computational Limits of Modern Hopfield Models: A Fine-Grained Complexity Analysis — **ICML 2024**
Jerry Yao-Chieh Hu*, **Thomas Lin***, Zhao Song, Han Liu. — ArXiv Link

ACHIEVEMENTS & LEADERSHIP

The Residency — *Resident (f26 cohort)* — San Francisco, USA | Fall 2026
- Selected for The Residency's founder cohort (20 of 2,000 applicants).

Conference Reviewer — 2026
- NeurIPS 2026, ICLR 2026 (AI & PDE Workshop), ICML 2026 (AI4Science Workshop).

Harvard Undergraduate in Asia Program (HUAP) NTU Branch – *PR Member* — Taipei, Taiwan | 2024–25
- Fundraised a record $4,500 (4 deals) — 97% of the 7-member PR team's total.
- Hosted Harvard undergraduate students in March 2025 for HUAP's Spring Conference at NTU.

European Physics Olympiad (EuPhO) – *Bronze Medalist* — Ljubljana, Slovenia | May 2022
- Fundraised $14,000 for Taiwan's olympiad participation from companies, foundations, and government.

SKILLS & INTERESTS

Research: Neural Operators, Generative Models, Inverse Problems, AI for Science, World Models, Control Theory
Technical: Python (PyTorch, Pandas, NumPy, scikit-learn, Statsmodels, Matplotlib, Hiplot), CUDA, Gurobi, MATLAB